

Mohammad Ashraf Mohammad Elsayed

AI & Machine Learning Engineer

mohammad.a.hammad.eng@gmail.com | +20 102 589 6994 | Cairo, Egypt | [GitHub](#) | [LinkedIn](#)

SUMMARY

AI & Machine Learning Engineer specializing in NLP, agentic RAG and multi-agent LLM systems, with a B.Sc. in Information Systems (GPA 3.50/4.0, Very Good with Honors). Built a self-improving clinical RAG assistant in LangGraph with hybrid retrieval and cross-encoder reranking that reached 92.5% precision@1 on a 40-question evaluation set, and a scalable RAG framework on FastAPI, Qdrant/pgvector and Docker with provider switching across OpenAI, Cohere and Google LLMs. Currently a Data Science and AI Engineer Intern at PM Accelerator (ML forecasting pipeline reaching 94% R²), following hands-on LLM agent training at Orange.

EXPERIENCE

Data Science and AI Engineer Intern - PM Accelerator, Remote (California, USA) Aug 2026 – Present

- Built an end-to-end ML forecasting pipeline (Linear Regression, Random Forest, XGBoost, Ensemble) on 40+ global weather features, achieving 94% R² for temperature and 68% R² for PM2.5 air-quality prediction, with modular preprocessing and feature-importance analysis to surface key climate drivers.
- Contributing to a production-ready AI product in a cross-functional team, owning model design and pipeline architecture (under NDA).

AI Agent Training - Orange Business Center (Cairo, Egypt) [demo link](#) Jul 2026 – Sep 2026

- Built LLM-powered AI agents with LangGraph, integrating tool calling, structured outputs, intent recognition and dynamic task execution.
- Developed Retrieval-Augmented Generation (RAG) pipelines that retrieve and generate answers from external knowledge sources, including documents and Excel data.
- Applied Transformer and LLM concepts, prompt engineering, output parsing, knowledge integration and model fine-tuning in practical Python implementations.

SKILLS

Languages & Data: Python, C++, SQL, Pandas, NumPy, Matplotlib

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, OpenCV, Supervised & Unsupervised Learning, CNNs, RNNs, Transformers, Attention Mechanisms, ViT, U-Net, SMOTE, Model Evaluation

GenAI & Agents: LLMs, LangChain, LangGraph, RAG, Agentic RAG, Multi-Agent LLM Pipelines, Prompt Engineering, LoRA Fine-Tuning, Tokenization, Hybrid Search, Cross-Encoder Reranking, Ragas

Databases & Vector Stores: PostgreSQL, pgvector, SQLAlchemy, Alembic, Qdrant, ChromaDB

Backend & MLOps: FastAPI, Docker, MLflow, DVC, CI/CD, Prometheus, Grafana, Streamlit, n8n, Git, GitHub

PROJECTS

[Clinical Guidelines RAG: Agentic Clinical Assistant & API Platform](#) (Hackathon Project)

- Built a self-improving agentic RAG pipeline (LangGraph) with hybrid dense/sparse retrieval (RRF) and cross-encoder reranking; achieved 92.5% precision@1, 100% source-document retrieval accuracy and 86.4% concept coverage on a 40-question clinical evaluation set, with 97.5% of retrievals passing LLM relevance grading in ≤1 iteration on average.
- Designed a role-based multi-agent supervisor (doctor/patient) with bilingual Arabic/English symptom and vital-sign checkers, served through a FastAPI backend with async REST, SSE streaming and session management.
- Extended access beyond the web app with voice (STT/TTS via Colab, exposed through Cloudflare Tunnel) and a Telegram bot for text/voice interaction, including automated appointment scheduling via n8n.

[AdvancedRAG: Scalable Retrieval-Augmented Generation Framework](#) (Graduation Project)

- Built a scalable, modular RAG framework with FastAPI and a provider factory for switching between OpenAI, Cohere and Google LLMs and between vector stores (Qdrant, pgvector).
- Engineered the retrieval pipeline with token-aware chunking, semantic boundary detection, hybrid search (vector + keyword + metadata) and semantic caching for repeated queries.
- Designed production-ready infrastructure: a PostgreSQL schema with SQLAlchemy/Alembic and a fully Dockerized deployment for reproducibility.

Credit Card Fraud Detection

May 2025

- Built an ML pipeline to detect fraudulent transactions under severe class imbalance (305 minority vs. 170,579 majority instances), comparing random undersampling, random oversampling, combined over/undersampling and SMOTE to handle the imbalance.
- Benchmarked Logistic Regression, KNN, PCA + KNN, Neural Networks and Random Forest; the best model (Random Forest) reached an F1 score of 0.88, recall of 0.84 and precision of 0.92.
- Ran a precision-recall threshold analysis, tuning the decision threshold to 0.56 to maximize precision (92.6%) while keeping recall at or above 80%.

EDUCATION

Bachelor of Information Systems and Technology, Port Said University | GPA: 3.50/4.0 (Very Good with Honors) Oct 2022 – Jul 2026

CERTIFICATES

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- CSkilled Academy: Machine Learning Diploma (140 hours), taught by a senior AI expert at NVIDIA with 10+ years at the company | Jan 2025 – Jul 2025 | [Certificate](#)

ACHIEVEMENTS

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- AI Hackathon: Top 3 of 20 teams | [Certificate](#)

LANGUAGES

Arabic (Native), English (Professional working proficiency)